Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

Weekly Progress Report — Week 9

Aryan Thakkar · **Nikkesh Parekh** · **Hitaksh Thaker** · **Maitri Parekh**

ECE 501 – Digital Image Processing Course Project

Faculty: Prof. Mehul Raval · **TAs:** Apoorva Mehta · Jyoti Triklani · Sagar Chavda

Date: November 14, 2026

1 Purpose and Scope

Week 8 generalized the pipeline to several targets but scored only Stephan's Quintet. Week 9 is the final week, so its goal was to turn the pipeline into a complete, evaluated system: measure standard morphology statistics on every available Hickson Compact Group (HCG) image, classify with a fixed rule, and score the result against published morphologies. The main items are:

1. **Shared engine and measurement**: one script (`galaxy_pipeline.py`) used by both the notebook and a command-line runner. It segments each image and measures Concentration, Asymmetry, Smoothness (CAS), Gini and M_{20} on linear flux (Section 2).
2. **Ground truth and data growth**: a labelled catalogue of Hickson members, and a systematic search of the HST archive to enlarge the sample from 9 to 26 fields (Section 3).
3. **Fixed rule and honest scoring**: a human-readable threshold rule (E / S / Irr), scored by leave-one-group-out cross-validation (LOGO-CV) and by frozen-rule hold-out tests (Section 4).
4. **Robustness experiments**: neighbour-model asymmetry, band calibration, truncation handling, disc detection, a bulge gate and nested model selection. Most were negative results and were not adopted (Section 5).

No machine learning is used anywhere. The classifier is a fixed threshold rule. To avoid optimistic scores, no setting was tuned on hold-out groups and no per-image parameters were added. Data were added only through group-independent rules.

2 Pipeline Engine and Measurements

2.1 Methodology

For each FITS image the engine estimates the sky and a no-data mask (non-finite and low-weight pixels, carried over from the Week 8 HCG 16 fix), then segments sources using a significance threshold. Galaxy positions are matched to catalogue entries by WCS, with a default match radius of $10''$. For every matched, measurable galaxy it computes:

- **C**: concentration $5\log_{10}(r_{80}/r_{20})$ within the Petrosian aperture.
- **A**: rotational asymmetry, background-corrected.
- **S**, **Gini**, $\mathbf{M_{20}}$: smoothness and light-distribution statistics.

Everything is measured on linear flux, not on stretched images. A galaxy counts as measurable only if its signal-to-noise per pixel is adequate and its aperture is not truncated. This last condition (`max_clipped_frac`$= 0.15$) comes from a simulation described in Section 5.

2.2 Scaling and Verification

The engine gained a process pool (`--workers N`). All 42 image jobs finish in about 6 minutes on 8 workers, against about 80 minutes serially, with identical output to the serial run. A reload function re-classifies a finished run without reprocessing, and a per-image audit reports each labelled member as matched, measurable, or lost for a stated reason. In the audit, 88 labelled member–image pairs fell on valid pixels, 84 were matched and 80 were measurable. The misses have physical causes: a frame-edge truncation (HCG 96b), faint knots inside a larger galaxy (HCG 18c/d), and low signal-to-noise (HCG 22c, 31d). One true segmentation failure remains, an edge-on Sdm (HCG 79d) merged into its bright neighbour.

3 Ground Truth and Data Growth

3.1 Labels

Ground truth is the Hickson (1989) morphology table from VizieR (VII/213), mapped to three classes before any evaluation: E/S0 $\rightarrow$ E; Sa–Sd $\rightarrow$ S; Sdm/Sm/Im $\rightarrow$ Irr. S0/a is left unlabelled. Stephan's Quintet keeps its

NED/Galaxy Zoo labels. Hickson types were cross-checked against HyperLEDA: 69 agree, 3 could not be verified and 1 is disputed (HCG 95c). A HyperLEDA catalogue was also built with a fixed rule (T-type error ≤ 1.5, positions excluded if within $5''$ of a Hickson entry). It added only one new label (IC 399), because background galaxies in these fields are too faint to have published types.

3.2 Archive survey

An archive survey script queries MAST for each group and selects images by a fixed priority: ACS and WFC3 optical bands first, then WFPC2, with WFC3/IR used only where no optical band covers a member. A greedy set-cover picks several pointings per group when needed. While doing this we found a bug: MAST names WFPC2 products `WFPC2/PC` and `WFPC2/WFC`, so the old exact-match test never selected a WFPC2 image. After the fix and parallel queries (about 3 minutes instead of about 70), the survey found only 93 labelled members in 31 groups appear in *any* calibrated HST image. The archive, not the method, limits the sample. The final data set is 58 images, 42 jobs and 26 fields, with 73 labelled galaxies measurable in 26 groups (63 after the truncation rule).

4 Classification Rule and Evaluation

4.1 Rule and protocol

The rule uses only C and A and is read in this order: **Irr** if $A \geq A_{\text{Irr}}$; otherwise **E** if $C > C_E$ and $A < A_E$; otherwise **S**. The thresholds are fitted by a grid search that maximises accuracy. Headline accuracy is LOGO-CV: each fold holds out an entire group, fits on the rest and predicts the held-out group, so a group's galaxies never help classify one another. Resubstitution accuracy is shown for reference only. We also froze the rule before new data were downloaded and applied it unchanged to galaxies it had never seen. These hold-out scores are the most reliable numbers here.

Table 1: Main results. Majority-class baseline for the hold-out was 57.1%.

Test	Galaxies	Accuracy
Rule fitted on first 18 groups, LOGO-CV (start of this period)	52	73.1%
Same rule frozen, applied to unseen galaxies (new groups)	21	76.2% (16/21)
Frozen rule on 8 further unseen groups after a second archive pass	14	78.6% (11/14)
Final rule, 3-class LOGO-CV	63	**77.8%** (49/63)
Final rule, E vs. S only	51	**86.3%**
Final rule, early vs. late type	63	**88.9%**

Final thresholds: Irr if $A \geq 0.317$; E if $C > 3.653$ and $A < 0.286$; else S. Balanced accuracy for the 3-class case is 77.4%. If the 14 galaxies dropped as unmeasurable are counted as errors, the figure is 49/77 = 63.6%, and a nested version of the rule selection gives 71.4%. We report these lower figures as well because the 77.8% depends on which galaxies are measurable.

Early vs. late type pools E/S0 as early and S + Irr as late, using an asymmetry-only threshold. E vs. S drops the Irr galaxies (51 remain). The cross-validated scores above are computed on the same 63 galaxies used to design the pipeline, so they are somewhat optimistic (the nested 3-class score is 71.4%). The frozen hold-out scores on unseen galaxies are the unbiased estimate.

4.2 Sources of variance

The score depends on which groups are in the sample: an earlier frozen rule fitted on 9 groups scored only 46% on 9 unseen groups. The same code and configuration on a Mac and on the Windows machine differ by about 5 points (69.9% vs. 64.4% on the same 73 galaxies) because 11 crowded galaxies segment differently across library builds. Differences of this size between variants are therefore measurement noise.

5 Robustness Experiments

Each idea below was written down with its acceptance rule before it was evaluated, and was adopted only if it improved the cross-validated score on every task.

Table 2: Experiments after the baseline rule. "Adopted" means it changed the production pipeline.

Idea	Result	Adopted
Neighbour-model asymmetry	Subtract smooth elliptical models of all other objects before measuring A. Lowers A for contaminated E galaxies (HCG 79c 0.25→0.12, 68b 0.31→0.08) but raises A for spirals, so the E/S gap shrinks. LOGO-CV 69.9%→64.4%.	No
Band calibration	Map F606W to F814W with a label-free fit ($A_{814} = 0.897A_{606} - 0.022$, 28 objects). No consistent gain.	No
Truncation rule	Simulation shows A is biased by > 0.03 beyond about 17–21% clipped aperture, so galaxies above 15% are not measured. Recovering the aperture failed its own test.	**Yes**
Disc detection	Isophote ellipticity finds discs correctly (spirals 0.51 vs. E 0.22) and fixes HCG 7a, 22b, but mislabels inclined S0s (HCG 56b/c, 79b). 76.2%, E vs. S tie.	No
Bulge gate on Irr	Irr only if $A \geq A_{hi}$ and $C < C_I$. Fixes HCG 16a, breaks four others. 73.0% vs. 77.8%.	No
Other feature sets	CAS, Gini–M_{20}, residual-flux fraction, A inside $0.5\,r_p$: none beat C+A under nested CV (+1 galaxy at most).	No

5.1 Error analysis

The remaining errors fall in a few classes. Crowded or tidally disturbed ellipticals (HCG 40b, 68a, 79b/c, 95a) have high A and are called S, and much of this asymmetry appears to be real tidal distortion, since the neighbour models did not remove it. Spirals imaged only in F606W look clumpier and are called Irr (HCG 40c/d, 16a, 44a). Bulge-dominated spirals have high C and are called E (HCG 7a, 96c, 22b). Frame-edge truncation affects HCG 46b, 68a and 68b, and HCG 16d is probably a label error. Three of the twelve Irr galaxies (HCG 16c, 95c, 31c) are concentrated starbursts, so C cannot separate Irr from disturbed spirals in compact groups. Within the E class the S0 galaxies carry the asymmetry (median A 0.18 against 0.03 for true E), so the main E/S confusion is S0 against Sa.

6 Conclusion

The project now has a complete, evaluated classical pipeline: 42 image jobs run in minutes, measurements are validated by a per-image audit, and a two-feature threshold rule reaches 77.8% three-class LOGO-CV, 86.3% for E vs. S and 88.9% for early vs. late type. On galaxies it had never seen, the frozen rule scored 76.2% and 78.6%, well above its earlier 46%. The caveats are the small sample (63 measurable galaxies in 26 groups), a ± 5 point sensitivity to the software build, and the fact that only about a third of the labelled members exist in the HST archive at all. Most of the extra ideas tested did not help, and we report them as negative results rather than tuning them in. Further work would be a spiral-arm measure on the residual after subtracting the isophote model, colour where several bands exist, and a larger archive sample including JWST.

End Of Weekly Report